

Chapter 187

Rendering Media

This section describes the options that are available for file-based delivery.

The workflow is simple; you define the format and other settings that dictate how the media is to be rendered, define a range of clips in the currently selected session, and then add a job containing these settings to the Render Queue.

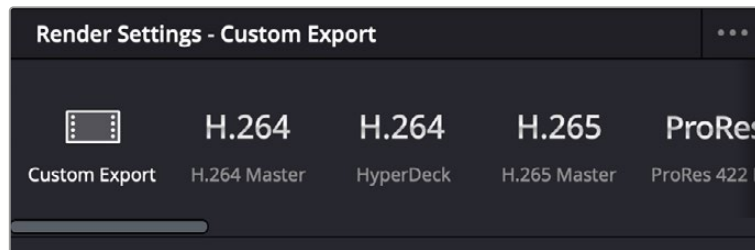
You can queue up as many different render jobs as you like, each with different formats, output options, and ranges of clips, depending on what you're trying to accomplish. When you're ready to render, simply click the Start Render button.

Contents

Using Presets for Fast Rendering	4034	Video Panel.....	4046
Custom.....	4034	Audio Panel.....	4054
Social Media Presets.....	4034	File.....	4056
ProRes Master.....	4038	Additional Outputs	4058
H.264 Master.....	4038	How to Avoid Overwriting Clips	
H.265 Master.....	4039	When Rendering Output Media	4059
IMF (Studio Version Only).....	4039	Defining a Range of Clips	
Frame.io.....	4039	and Versions to Render	4059
Final Cut Pro 7 or X XML.....	4040	Choosing Which Versions	
Premiere XML.....	4041	to Render for Each Clip.....	4060
Avid AAF.....	4041	Using the Render Queue	4061
Pro Tools.....	4041	Rendering Jobs from	
Audio Only.....	4043	Multiple Projects at Once	4063
Creating and Using Your Own Presets.....	4043	Remote Rendering	4064
Import and Export Custom Presets	4043	Using Multiple Project	
Choosing a Location to Render	4045	Libraries in Remote Rendering.....	4064
Single Clip vs. Individual Clips	4045	Sharing Storage.....	4064
Single Clip.....	4045	Setting Up and Using Remote Rendering.....	4064
Individual Clips.....	4046	When You're Finished Remote Rendering.....	4065
All Other Render Settings for Output	4046	Setting Up a "Headless"	
		Remote Rendering Workstation.....	4065

Using Presets for Fast Rendering

The very top of the Render Settings list has a set of presets for many of the most common rendering workflows you'll need to accomplish. If you want to create your very own settings, then choose custom. Each preset automatically sets up what you need and locks you out of settings that are not necessary for rendering that type of media.



Render presets selection

Custom

When custom is selected, nothing is automatically set, and all conventional media rendering options are available, except for those that are specifically associated with particular presets. You must manually choose the settings and options you need. All Render settings are saved on a per-project basis.

Social Media Presets

These presets let you render media specifically for video sharing services, with the option to upload the rendered files automatically.

YouTube 720p/1080p/1440p/2160p

A dropdown menu lets you choose four different resolutions to render to and selects the appropriate settings for exporting your program as a file suitable for uploading to YouTube and many other video file sharing services. The preset renders a single clip at the timeline's frame rate and sets the following parameters:

- **Format:** MP4
- **Video Codec:** H.264
- **Encoding Profile:** High
- **Audio:** Main 1
- **Audio Codec:** AAC
- **Data burn-in:** Same as Project
- **Normalize Audio:** Checking this box will normalize the clip's audio level to YouTube's standard.
- **Use Proxy Media:** Lets your final render use the proxy media instead of the original source camera files. Useful if speed is more important than quality.
- **Render without Timecode:** If you're using certain editing tools on mobile devices that have issues with timecode, checking this box lets you Render without Timecode.

- **Upload directly to YouTube:** When this box is checked, the resulting render will automatically upload to your YouTube account, and the following parameters will become available. If you have multiple YouTube Channels on your account, you can select which channel to upload to.

Title: Enter the title of your video.

Description: Enter a description of your video.

Chapters from Markers: Checking this box embeds chapter points in the YouTube video corresponding to the the selected marker color and the marker's position on the Timeline.

Upload Thumbnail: Checking this box will open a file browser, allowing you to select a still image that will become the thumbnail image for your video on YouTube.

Visibility: You can change how your video will be accessible on YouTube.

Category: You can select your video's YouTube category.

Vimeo 720p/1080p/2160p

A dropdown menu lets you choose three different resolutions to render to and selects the appropriate settings for exporting your program as a file suitable for uploading to Vimeo and many other video file sharing services. The preset renders a single clip at the timeline's frame rate and sets the following parameters:

- **Format:** MP4
- **Video Codec:** H.264
- **Encoding Profile:** Auto
- **Audio:** Bus 1
- **Audio Codec:** AAC
- **Data burn-in:** Same as Project
- **Use Proxy Media:** Lets your final render use the proxy media instead of the original source camera files. Useful if speed is more important than quality.
- **Upload directly to Vimeo:** When this box is checked, the resulting render will automatically upload to your Vimeo account, and the following parameters will become available.

Title: Enter the title of your video.

Description: Enter a description of your video.

Visible To: You can change how your video will be accessible on Vimeo, including password protection.

TikTok Deliver Page Preset

A dropdown menu lets you choose two different resolutions to render to and selects the appropriate settings for exporting your program as a file suitable for uploading to TikTok and many other video file sharing services. The preset renders a single clip and sets the following parameters:

- **Resolution:** 1920x1080 HD
- **Use Vertical Resolution:** Unchecked. Check this box if you want to deliver your video in portrait mode for proper display on phones. This should be on for TikTok.
- **Frame rate:** The chosen frame rate of your timeline. You can also override this and set another frame rate manually.
- **Format:** MP4
- **Video Codec:** H.264
- **Encoding Profile:** Auto
- **Audio:** Bus 1
- **Audio Codec:** AAC

- **Data burn-in:** Same as Project
- **Use Proxy Media:** Lets your final render use the proxy media instead of the original source camera files. Useful if speed is more important than quality.
- **Upload directly to TikTok:** When this box is checked, the resulting render will automatically upload to your TikTok account, and the following parameters will become available.

Title: Enter the title of your video.

Visible To: Lets you choose who will be able to view this video. The options are Private, Public, and Friends.

Allow comments: Checking this box allows commenting on your TikTok video. Un-checking this box forbids comments on your video.

Allow Duet: Checking this box will allow your video to be used side-by-side with a video from another creator in TikTok using the Duet function.

Allow Stitch: Checking this box will allow your video to be edited and combined with a video from another creator in TikTok using the Stitch function.

Presentations Preset

This setting lets you flag and format timelines to be used with Presentations in Blackmagic Cloud. Unlike other presets, there are no codec settings to choose from. This setting simply adds the metadata necessary for Presentations to share your timeline. You must have already created a Presentation in Blackmagic Cloud before you can use this setting.

- **File Name:** Choose the name of the timeline to be uploaded to Presentations.
- **Upload To:** Choose the name of the existing Blackmagic Cloud Presentation to upload to.
- **Use Proxy Media:** Use the Proxy Media as the source for the compression and upload to Blackmagic Cloud rather than the original media.

Dropbox or Dropbox Replay 720p/1080p/2160p

A dropdown menu lets you choose three different resolutions to render to and selects the appropriate settings for exporting your program as a file suitable for uploading to Dropbox and many other video file sharing services.

The preset renders a single clip at the timeline's frame rate and sets the following parameters:

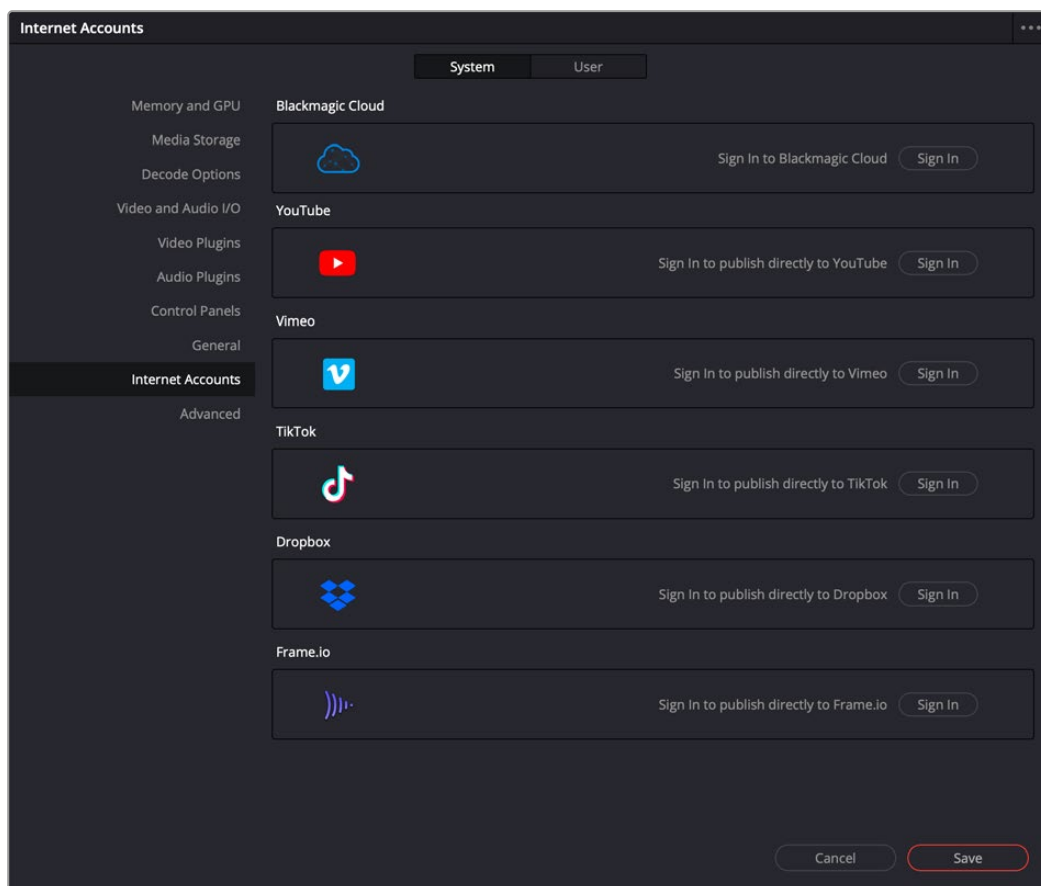
- **Format:** MP4
- **Video Codec:** H.264
- **Encoding Profile:** Auto
- **Audio:** Bus 1
- **Audio Codec:** AAC
- **Data burn-in:** Same as Project
- **Use Proxy Media:** Lets your final render use the proxy media instead of the original source camera files. Useful if speed is more important than quality.
- **Upload directly to Dropbox:** When this box is checked, the resulting render will automatically upload to your Dropbox account.

Setting Up Video Sharing Uploads

DaVinci Resolve has account integration with YouTube, Vimeo, TikTok, Dropbox, and Frame.io that allows you to render and upload directly to each service. An Internet Accounts panel in the System tab of the DaVinci Resolve Preferences lets you sign into your YouTube, Vimeo, Dropbox, and Frame.io accounts, as well as specify a local cache location for media being synced with Frame.io.

For each service you sign into, a floating window presents the interface in which you'll need to enter your login name and password to enable integration, followed by whatever two-factor identification and other required steps are necessary. Once entered, DaVinci Resolve will sign in to each of these services automatically when DaVinci Resolve opens.

For each service you sign into, the service's upload parameters will also be available in the Custom Export Video settings in the Deliver page. This allows you to create your own custom file settings that can supersede the normal formats allowed by the service presets, while allowing you the convenience of the all-in-one uploading and description entry of the preset.



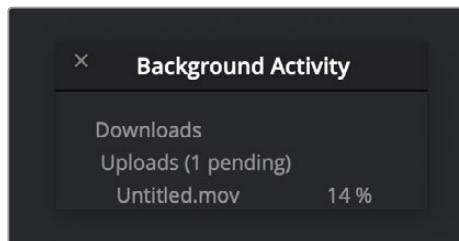
The Internet Accounts panel of the System tab of the DaVinci Resolve Preferences window

NOTE: For Frame.io, the local cache location is used to store clips you import into a DaVinci Resolve project from the Frame.io volume in the Media Storage panel of the Media page.

Deliver and Upload to Social Media

When you've configured YouTube, Vimeo, TikTok or Dropbox access in the Internet Accounts panel of the System Preferences, the YouTube, Vimeo, and Dropbox presets expose an "Upload directly to YouTube/Vimeo/TikTok/Dropbox" checkbox, which lets you choose whether or not to automatically upload the rendered result.

Choose the desired export options, then click the Add to Render Queue button to add this job to the Render Queue as you would with any other export. When that job is rendered, it automatically proceeds to upload to the selected video sharing service, and an upload percentage indicator appears in the job listing to show how far along this upload is. This upload is done in the background, so you can continue working on other things in DaVinci Resolve while the file uploads. If you want to see how long the upload will take on any other page, you can choose Workspace > Background Activity to see the Background Activity window.



Waiting for your movie to upload

Once the upload is finished, you can view the file directly on the social media site in a web browser by right-clicking on the job in the Render Queue and selecting Reveal in Browser.

Review Before Upload

For an extra level of quality control before you send your video out live on the internet, you can select Review Before Upload from the Render Queue option menu. This option will pause the automatic upload process after the file has been rendered with a message, "Waiting for Upload." You can then right-click on the job in the Render Queue, and select Reveal in Finder to open and review your video. Once you've made your decision, from that same menu, you can select either Upload to (service name) to let your video go, or Cancel Upload if you've had second thoughts.

ProRes Master

For quickly outputting ProRes Master files of a whole program. When selected, defaults to rendering in single clip mode, with the Format set to QuickTime, the Codec set to Apple ProRes, and the Type set to Apple ProRes 422 HQ. Audio defaults to the Codec being Linear PCM and the Bit Depth being 16.

H.264 Master

For outputting H.264 files of a whole program. When selected, defaults to rendering in single clip mode, with the Format set to QuickTime, and the Codec set to H.264. Quality, Encoding Profile, and Entropy Mode are set to Auto, Passes defaults to Single, and Key Frames default to Automatic with Frame Reordering turned on. Audio defaults to the Codec being AAC with the Data Rate set to 320 Kb/s and the Bit Depth set to 16.

H.265 Master

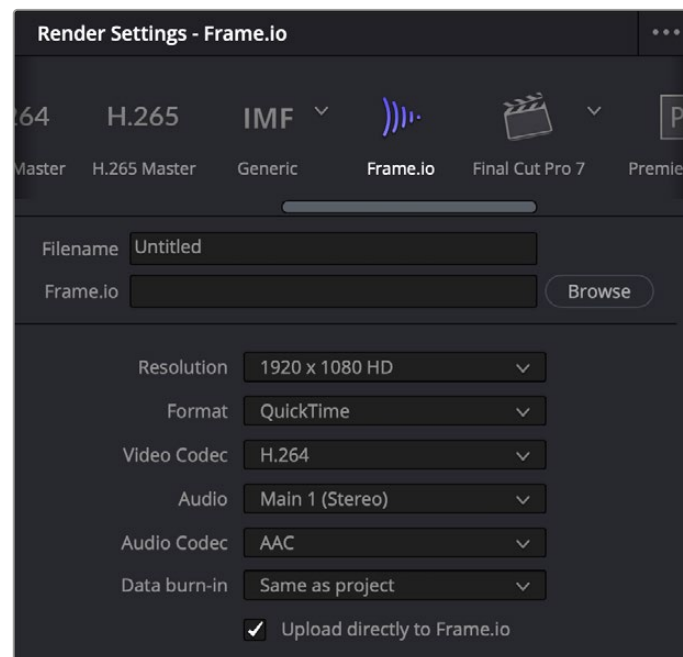
For outputting H.265 files of a whole program. When selected, defaults to rendering in single clip mode, with the Format set to QuickTime, and the Codec set to H.265. Quality is set to Automatic, Encoding Profile is set to Main, Passes defaults to Single, and Key Frames default to Automatic with Frame Reordering turned on. Audio defaults to the Codec being AAC with the Data Rate set to 320 Kb/s and the Bit Depth set to 16.

IMF (Studio Version Only)

A dropdown menu to the right of this preset provides options for Generic, 20th Century Fox, and Netflix-qualified presets. This preset is for facilities that deliver IMF files as digital-only deliverables. A Preset Type dropdown lets you choose the appropriate settings to populate the various locked IMF-specific parameters that appear.

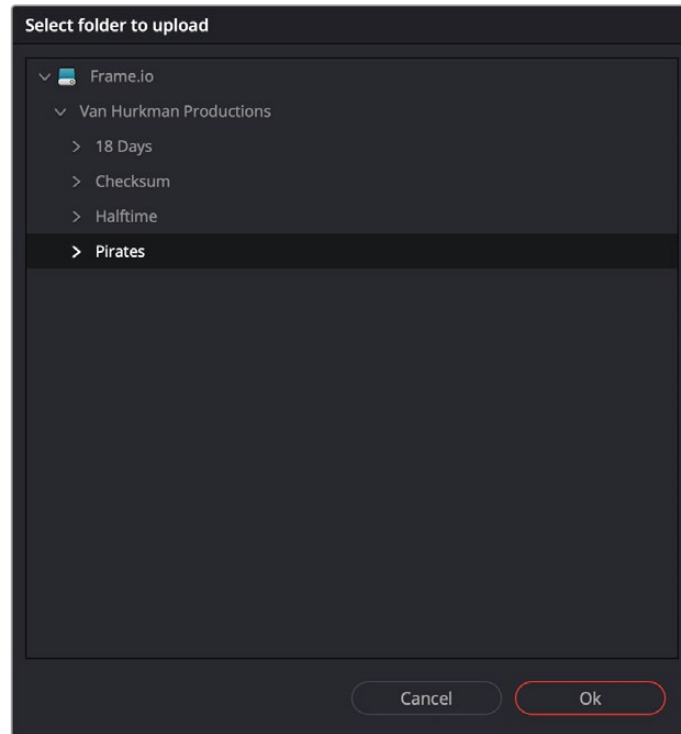
Frame.io

A Frame.io preset at the top of the Deliver page's Render Settings panel lets you render and upload a program for review. All options in the Render Settings panel update to present suitable controls for this process. At the bottom of the Render Settings list, an "Upload directly to Frame.io" checkbox lets you choose whether or not to upload the rendered result. A Description field lets you add a description to be uploaded along with the rendered media.



Choosing the Frame.io preset

When you choose the Frame.io preset, the Location field turns into a Frame.io field, and the Browse button lets you choose a project and folder path to which to upload the exported result.



Choosing a Frame.io account to deliver a program to

When you export to Frame.io, the available choices in the Resolution, Format, Video Codec, and Type dropdown menus are limited to those that are most suitable for Frame.io file sharing. Choose the desired export options, then click the Add to Render Queue button to add this job to the Render Queue as you would with any other export. When that job is rendered, it automatically proceeds to upload to Frame.io, and an upload percentage indicator appears in the job listing to show how far along this upload is. This upload is done in the background, so you can continue working on other things in DaVinci Resolve while the file uploads. If you want to see how long the upload will take on any other page, you can choose Workspace > Background Activity to see the Background Activity window.

For more information about Frame.io integration, see *Chapter 13, “Frame.io and Dropbox Replay Integration.”*

Final Cut Pro 7 or X XML

A dropdown menu attached to this preset lets you choose from two different XML formats to be exported along with the media you’re rendering:

- Selects the appropriate settings for projects that were sent from Final Cut Pro 7 to DaVinci Resolve using XML. This is meant for situations when you’re rendering media intended for a return trip to Final Cut Pro (by exporting an XML file from the Edit page). Renders Individual Clips, the “Codec” setting on macOS, defaults to Apple ProRes 422 (HQ), Output Size defaults to the current Timeline Resolution (as set in the Master Settings panel of the Project Settings), and Use Unique Filenames is turned on.

When you choose this preset, an XML of the timeline is automatically exported along with the media, with path names that reflect the rendered clips.

- Selects the appropriate settings for projects that were sent from Final Cut Pro X to DaVinci Resolve using XML. This is meant for situations when you're rendering media intended for a return trip to Final Cut Pro X (by exporting an FCPXML file from the Edit page). Renders Individual Clips, the "Codec" setting on macOS, defaults to Apple ProRes 422 (HQ), Output Size defaults to the current Timeline Resolution (as set in the Master Settings panel of the Project Settings), and Use Unique Filenames is turned on.

When you choose this preset, an XML of the timeline is automatically exported along with the media, with path names that reflect the rendered clips.

Premiere XML

Selects the appropriate settings for projects that were sent from Premiere Pro to DaVinci Resolve using XML. This is meant for situations when you're rendering media intended for a return trip to Premiere Pro. Renders Individual Clips, the "Codec" setting on macOS defaults to Apple ProRes 422 (HQ), Output Size defaults to the current Timeline Resolution (as set in the Master Settings panel of the Project Settings), and Use Unique Filenames is turned on.

When you choose this preset, an XML of the rendered timeline is automatically exported along with the media, with path names that reflect the rendered clips.

Avid AAF

Selects the appropriate settings for projects that were sent from Avid Media Composer or Symphony to DaVinci Resolve using AAF. This setting is NOT for exporting to Pro Tools. This is meant for situations when you're rendering media intended for a return trip to Media Composer (by exporting an AAF file from the Edit page). The "Codec" setting defaults to DNxHR 444 12 bit, Output Size defaults to the current Timeline Resolution (as set in the Master Settings panel of the Project Settings), and Render Clip with Unique Filename is turned on.

When you choose this preset, an AAF of the timeline is automatically exported along with the media, with path names that reflect the rendered clips.

Pro Tools

As of DaVinci Resolve version 16, Pro Tools export has been dramatically improved. This preset presents the appropriate options for exporting a specifically formatted AAF project file, linked audio files, and a linked reference video file to Pro Tools, or any application capable of importing a Pro Tools formatted AAF file.

When exporting using the Pro Tools preset, you must use the AAF file that's automatically created and written to the target location, because it's formatted specially for Pro Tools and it contains path names reflecting the rendered clips. Do not export an AAF using the File > Export AAF/XML command, as this will not provide the correct exchange file for Pro Tools, and it won't work correctly.

When you use the Pro Tools preset, DaVinci Resolve outputs the following:

- 1 What you choose in the Codec dropdown menu of the Audio panel dictates whether you export the audio from the Timeline as a collection of files that link to a separate AAF, or an AAF with audio file embedded within as a single deliverable.
 - Choose Linear PCM to export individual files linked to a separate AAF interchange file
 - Choose Embedded in AAF to export an AAF with embedded Broadcast WAV audio files within it as a single deliverable

Whether you export separate files or a single embedded AAF deliverable, each of the audio clips in the current Timeline can be exported as individual mono or multichannel audio files. The standard mono round trip export from DaVinci Resolve to Pro Tools is the default setting, with

the “Render one track per channel” box checked in the Audio tab of the Pro Tools Render Setting. With this option, a 5.1 polyphonic .wav file would be exported as six individual mono .wav files.

If the “Render one track per channel” box is unchecked, DaVinci Resolve will output multichannel polyphonic .wav files instead. If you do this, it’s important to check in advance that Pro Tools supports the particular multi-channel formats you want to export before committing to this workflow.

Each exported file contains every audio channel from the source media, regardless of channels that have been muted in the audio panel of Clip Attributes. This means no matter how the video editor organized the channels of audio in the Timeline, you’ll always deliver every channel of each audio clip to whomever is doing your audio postproduction.

- 2 You can also choose to include handles using the “Add X frame handles” option in the Advanced Settings of the Video panel to add extra frames to the beginning and end of each exported audio clip. This will provide needed editing flexibility to whomever is refining your audio.
- 3 The type of audio file that’s exported is determined by your choice of video format in the Video panel:
 - If you choose the MXF OP-Atom video format, then MXF audio files will be exported.
 - If you choose the QuickTime format, then Broadcast Wave files will be exported.
- 4 All video in your timeline will be rendered and output as a single reference movie, in the format that’s selected in the Video panel, with all effects and titles baked in. Subtitles can also be burned into the reference movie or exported as a file. If you want to provide a window burn, you can enable visible metadata using the Workspace > Data Burn-In window. If you do not wish to export a reference movie, you can uncheck the Export Video box in the Video panel.

When you output using the Pro Tools preset, an AAF of the audio tracks of the current Timeline is exported that’s formatted for import into Pro Tools, or any other digital audio workstation (DAW) software that’s compatible with the Pro Tools style of AAF import.

- Exported audio files have the file name and timecode of the source media they were extracted from, to enable relinking to the source media in Pro Tools, if necessary. In the case of Video+Audio files that have been synced in DaVinci Resolve, exported audio files are given the timecode and name of the synced audio source file, not that of the video clip.
- Each audio track exports whatever custom name you may have given it, for use by Pro Tools.
- All track and clip volume automation is exported, with all keyframes.
- iXML metadata is also exported, including channel names when available.

IMPORTANT: When you export to Pro Tools in the Deliver page, audio effects are neither exported nor baked in, which means that FairlightFX, EQ, Compression, Pitch, and Elastic Wave effects will be ignored. If you are experiencing problems with imported AAF files, check to see if there are audio effects or audio compound clips in the Timeline, and replace any you find with duplicates of the same audio clips that have no effects.

Audio Only

This preset is specifically for rendering an audio-only media file from the Timeline. Video rendering is disabled, and this preset defaults to rendering the Main 1 bus as a single clip, rendering one track per channel using the MXF OP-Atom format set to the Linear PCM codec, at 16-bits. However, the QuickTime, MP4, and WAV formats are also available, and you can also render 24- or 32-bit output. Additionally, you have the option to render other Mains or Submixes, or to choose a specific Timeline Track to render. Finally, you can choose to render the current program as Individual Clips.

Creating and Using Your Own Presets

If there is a particular group of settings that you find yourself using repeatedly, you can turn it into a custom Easy Setup, for easy recall.

To create a new Easy Setup:

- 1 If you want to start from scratch, make sure to choose Custom from the preset panel to unlock every setting in the Render Settings pane.
- 2 Choose the particular settings you require in the Video, Audio, and File panels for your new preset.
- 3 Open the Render Settings Options menu, and choose Save as New Preset.
- 4 Type a name into the “Render Preset” dialog, and click Save. The new preset now appears in the Preset panel.

To load a preset:

- Click any preset. Every setting in the Render Settings pane updates to reflect the preset you selected.

To change a custom preset that you’ve created

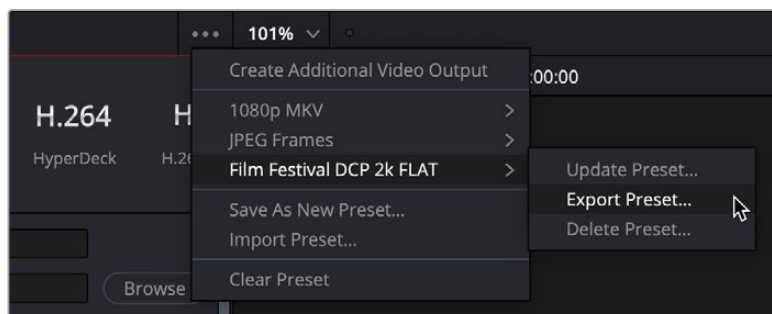
- Click a preset you want to change, make whatever changes you need to in the Video, Audio, and File panels, then click the Render Settings Option menu, and choose Update Current Preset.

To delete a custom preset that you’ve created:

- Click a preset you want to delete, then click the Render Settings Option menu, and choose Delete Current Preset.

Import and Export Custom Presets

Custom render presets that you create can be imported and exported from the Deliver page. Presets are saved in .xml files that can be easily sent to other users or workstations to ensure the exact same delivery methods are used between machines. In addition, any newly created export presets are available in the “Add to Render Queue Using” menu when you right-click a timeline.



Render presets can now be imported and exported from the Deliver page.

To save a custom render preset:

- 1 In the Deliver page, adjust the Video, Audio, and File options in the Render Settings to your needs.
- 2 In the Render Settings Option (3-dot) menu, select Save as New Preset.
- 3 Enter a name for the new preset, and press OK.

The new preset will be available in the Render Settings Option menu, as well as an option in the Custom Preset Settings icon in the Render Presets row at the top of the Render Settings.

To export a custom render preset:

- 1 In the Deliver page, click on the Render Settings Option (3-dot) menu.
- 2 Select the name of the saved preset you wish to export.
- 3 Select Export Preset from the dropdown menu.
- 4 Use the file browser to name and set the save location of the preset.

The new preset will be available as an .xml file.

To import a custom render preset:

- 1 In the Deliver page, click on the Render Settings Option (3-dot) menu.
- 2 Select Import Preset from the dropdown menu.
- 3 Select the Render Setting .xml file in your file browser.
- 4 Press Open.

The imported preset will be available in the Render Settings Option menu, as well as an option in the Custom Preset Settings icon in the Render Presets row at the top of the Render Settings.

To update a custom render preset with new settings:

- 1 Make any changes you wish to the Video, Audio or File Render Settings.
- 2 In the Deliver page, click on the Render Settings Option (3-dot) menu.
- 3 Select the name of the saved preset you wish to update.
- 4 Select Update Preset from the dropdown menu.
- 5 Press the Update Button from the warning dialog. This action can not be undone.

The selected preset will be updated with the new settings you selected.

To delete a custom render preset:

- 1 In the Deliver page, click on the Render Settings Option (3-dot) menu.
- 2 Select the name of the saved preset you wish to delete.
- 3 Select Delete Preset from the dropdown menu.
- 4 Press the Delete Button from the warning dialog. This action can not be undone.

The selected preset will be permanently deleted from the Render Presets list.

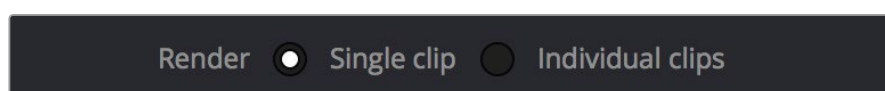
Choosing a Location to Render

The first decision you have to make when rendering your output is where it's going to be rendered. Accordingly, this is the first set of controls appearing at the top of the Render Settings parameters.

- **Filename:** A preview of what the file name will be based on the settings found in the File panel described later. The Custom/Timeline name and File suffix fields, as well as the “Use x digits in the filename” settings all determine what name appears here. The editable portions of this filename preview can also be edited here.
- **Location:** Click the Browse button to choose a directory in which to write the media being output by DaVinci Resolve. After you've selected a directory, the path name appears in the “Render job to” field.

Single Clip vs. Individual Clips

While there are numerous options available in the Render Settings of the Deliver page, there are basically two overarching ways you can render your project, depending on which of the “Render” radio buttons you click in the Output group.



Render a single clip or individual clips

Single Clip

When you select the Single clip option, you're setting up a render wherein all clips in the session are output together, as a single media file in whatever format you choose. This means you'll be rendering the selected range of the session to a single MXF or QuickTime file, or as a single collection of image sequences.

- **Timecode:** The timecode that's written out is dictated by the “Start timeline timecode at” setting of the timeline being rendered. Media files contain a continuous timecode track, while image sequences have timecode written into each frame's data header, and integrated into the file name (as a frame count).
- **Frame Rate:** If you're rendering a project that uses mixed frame rates, rendering to a single clip converts every clip in the entire session to the project frame rate, using either the project-wide or clip specific “Retime process” setting.
- **Effects:** Most effects are “baked into” the rendered output when you render a single clip.

IMPORTANT: Whenever clip filtering is enabled (via the dropdown menu to the right of the Clips button), Single Clip rendering cannot be selected. You can see if clip filtering is enabled by an orange line underneath the Clips button in the UI toolbar.

Individual Clips

Selecting the Individual clips option sets up a render where each clip is rendered as an individual media file in whichever format you choose. The result will be a collection of as many media files as there are clips in the range you've selected to render.

- **Timecode:** The timecode written to each clip is cloned from the original source media, making it easy to reconform media for projects being passed between DaVinci Resolve and NLEs.
- **Frame Rate:** If you're rendering a project that uses mixed frame rates, rendering to source renders each clip at its own individual frame rate, to accommodate round-trip workflows.
- **Effects:** You can choose whether any timeline-based effects are either ignored or "baked into" the individual clips by checking/unchecking the Render Timeline Effects box.
- **Resolution:** You can choose whether the individual clips are rendered at the timeline size or their original source resolution by checking/unchecking the Render at Source Resolution box.

All Other Render Settings for Output

This section covers the different render settings that are available for customizing your output. Depending on which Render Setting mode you chose, some of these may be hidden, but this section covers the full list found in the Advanced panel of controls.

If you choose one of the Easy Setups, then some of these settings will be locked, and others will be editable, depending on the requirements of that setup. If none of the Easy Setups is suitable for the task at hand, you can leave the Easy Setup dropdown menu set to none, and manually choose the necessary settings for the task at hand.

Video Panel

This panel contains all video-oriented parameters.

Format and Codec Controls

These top-level parameters let you choose whether or not to render video, and which format to render it to. Depending on which Format, Codec, and Type you choose, other options may or may not appear.

- **Export Video:** Turn this checkbox on to render the source video. Turn this checkbox off if you want to render the source audio all by itself; this disables all video controls, and shows an Audio Format dropdown menu in the audio section of settings.
- **Format:** A dropdown menu that gives access to the container formats that are currently available on your system. The available options depend on whether you have Final Cut Pro and QuickTime installed, and on the operating system you're using. This list is constantly growing, as new file formats are added over time, so be sure to check each new version for the latest supported formats.

AVI: A now-deprecated file-based media format that, despite its age, remains popular with Windows applications. Supports delivery using the Cineform, Grass Valley HQ and HQX, and Uncompressed RGB and YUV codecs.

Cineon: An older uncompressed image sequence format developed by Kodak, designed for film scanning and digital mastering, which delivers RGB 10-bit.

DCP: Native DCP encoding and decoding for creating unencoded DCP files only. If you have a license for Fraunhofer's EasyDCP, a setting in the Configuration panel of the System Preferences enables you to choose whether to use EasyDCP (for creating encrypted DCP output), or the native DaVinci Resolve encoding.

DPX: An uncompressed image sequence format favored by the film industry for mastering and delivery for DCDM mastering, which can be delivered as RGB 10-, 12-, 16-bit integer and half float, or RGBA 8-bit.

easyDCP: (when installed) An option that allows you to master a DCP or IMF directly from DaVinci Resolve in conjunction when you have an installed license of Fraunhofer's EasyDCP software.

EXR: The OpenEXR format is a high-dynamic-range image sequence format developed by ILM for applications requiring high quality and multiple channels. Used for outputting ACES and HDR deliverables. You can deliver to a variety of RGB Half and RGB Float settings. When choosing the RGB half (DWAA) or (DWAB) compression codecs, an additional "Compression level" setting appears that lets you choose how much compression to apply.

IMF: A native IMF encoding option that lets you export to the SMPTE ST.2067 Interoperable Master Format (IMF) for tapeless deliverables to networks and distributors, with support for encoding of JPEG2000 using a library licensed from Kakadu software. No additional licenses or plugins are required to output to IMF. The IMF format supports multiple tracks of video, multiple tracks of audio, and multiple subtitle and closed caption tracks, all of which are meant to accommodate multiple output formats and languages from a single deliverable. This is done by wrapping a timeline's different video and audio tracks (media essences) and subtitle tracks (data essences) into a "composition" within the Material eXchange Format (MXF).

JPEG 2000: DaVinci Resolve 15 introduced support for the encoding and decoding of JPEG2000 using a library licensed from Kakadu software. This includes a complete implementation of the JPEG2000 Part 1 standard, as well as much of Parts 2 and 3. JPEG2000 is commonly used for IMF and DCP workflows.

MJ2: The Motion JPEG 2000 format. DaVinci Resolve 15 introduced support for the encoding and decoding of JPEG2000 using a library licensed from Kakadu software. This includes a complete implementation of the JPEG2000 Part 1 standard, as well as much of Parts 2 and 3. JPEG2000 is commonly used for IMF and DCP workflows.

MP4: Dedicated MP4 encoding lets you export H.264-encoded movies.

MXF OP-Atom: A simple standard for the Material eXchange Format, a file-based media format, that's often used when delivering DNxHD. This version conforms to the SMPTE 390M standard, and can deliver using the DNxHD, DNxHR, Kakadu JPEG 2000, NTSC and PAL Avid, RGB Avid 10-bit, and XDCAM MPEG2 codec options.

MXF OP1A: A version of the Material eXchange Format that conforms to the SMPTE 378M standard, and can deliver using the 1080i Avid 8-bit, DNxHD, DNxHR, Kakadu JPEG 2000, NTSC and PAL Avid, RGB Avid 10-bit, Sony MPEG4 422 and 444, and Sony XAVC Intra CBG and VBR, and XDCAM MPEG2 codec options.

QuickTime: Apple's file-based media format, used when delivering Apple ProRes, DNxHD or DNxHR wrapped in QuickTime, GoPro Cineform RGB 16-bit and YUV 10-bit, Grass Valley HQ and HQX, Kakadu JPEG 2000, H.264, HEVC, H.265 (single or multi-pass), Photo JPEG, Kakadu JPEG 2000, Uncompressed 8- and 10-bit formats with ARGB/BGRA/RGB/YUV channel orders, and VP9 at 8-, 10-, and 12-bits.

TIFF: "Tagged Image File Format," an image sequence format compatible with many desktop video applications on many platforms and is also used when delivering for DCDM mastering.

- **Codec:** A dropdown menu that lets you choose from a selection of codecs that are available to the format you've selected above.
- **Type:** Different codec options may also present different bit depth and color space combinations, as well, which are available from this menu.

- **Maximum Bit Rate:** (Does not appear for all codecs) Codecs such as Kakadu JPEG 2000 let you specify a maximum bit rate, in Mbits per second, with which to encode the delivered video.
- **Field rendering:** If you're processing interlaced source material, this checkbox sets DaVinci Resolve to render each field individually before reintegrating them back into a single frame, in order to process clips most accurately with filtering operations that would otherwise violate field boundaries and cause problems. If you're not rendering interlaced media, you should leave this checkbox turned off, as it is more processing intensive.
- **Interlaced rendering:** By default, frame-based media sources (image sequences, etc.) are rendered progressively. You can now optionally render them as interlaced media instead by checking this box. Once checked, select the Field Order to encode, either top or bottom field first.
- **Export HDR10 Metadata:** (Available in Single clip mode if HDR10+ is enabled in Project Settings) Exports HDR10 metadata to the rendered file when you're doing an HDR workflow.
- **Embed HDR10 Metadata:** (Available in Single clip mode if HDR10+ is enabled in Project Settings) Exports HDR10 metadata to the rendered file when you're doing an HDR workflow. Embeds HDR10 metadata within the exported media of selected formats.
- **Embed HDR Vivid Metadata:** Exports HDR Vivid metadata to the rendered file when you're doing an HDR workflow. Embeds HDR Vivid metadata within the exported media of selected formats.
- **Render at Source Resolution:** (When rendering Individual Clips) This checkbox lets you render each clip at the same resolution as its source media file, letting you preserve mixed frame sizes for final delivery.
- **Resolution:** The output resolution for rendering. This setting defaults to Timeline Resolution; this lets you create a preset that automatically updates its resolution settings dynamically and automatically changed to the currently loaded timeline/project. However, you can change the resolution here if you want to set a specific resolution for the preset. Using this setting, you can queue up different render jobs at different resolutions, in order to output both HD and SD resolution media in the same render session, for example. Some file formats require specific resolutions, in which case the Output Size settings will be automatically set to the necessary resolution.
- **Frame rate:** (When rendering Single Clip) This setting defaults to the Timeline Frame Rate; this lets you create a preset that automatically updates its frame rate settings dynamically and automatically changed to the currently loaded timeline/project. However, you may wish to set this to a variation of the current conformed rate, for example choosing from between 23.98 or 24 fps. Doing so will adjust the metadata written within the file, which is used to aid playback for the range of systems available worldwide.

3:2 Pulldown Insertion Options: Starting with DaVinci Resolve Studio 12.5, you have the option of outputting either 29.97 or 30 fps media with 3:2 pulldown insertion if your project's playback frame rate is either 23.98 or 24 fps. To output 29.97 media, the project must be 23.98 fps; simply choose (23.976 3:2) from the Frame rate dropdown. Projects with 24 fps frame rates can only be output at 30 fps.

- **Chapters from Markers:** (QuickTime or MP4 only) Embeds chapter points in the rendered file corresponding to the marker's position on the Timeline of the selected marker color.
- **Export Alpha:** (When rendering Individual Clips) Turning this checkbox on results in alpha channels found in each clip's source media file being output to each delivered clip, as well as alpha information that you're creating in DaVinci Resolve and inserting into that clip via the Alpha output of the Color page Node Editor being output to each delivered clip.
- **Alpha Mode:** (When rendering Individual Clips) Lets you choose how to export alpha channels when Export Alpha is enabled. You can choose Straight or Premultiplied.

- **Render Stereoscopic 3D:** (Only appears if there are stereo clips in a timeline) Three options let you choose how to render stereoscopic timelines, rendering just one eye's worth of media at a time, or rendering a single set of stereo media in one of four ways, depending on the option you choose from the "Both eyes as" dropdown menu.

Left eye: Lets you render only the left-eye media from a stereo timeline.

Right eye: Lets you render only the right-eye media from a stereo timeline.

Both eyes as: Lets you select from four ways of rendering the left and right eyes of stereo media as a single set of media files. "Separate files" lets you output both the left-eye and right-eye media as individual media files, all at once. Side-by-side, Line-by-Line, and Top-Bottom let you output frame-compatible media that can be output to stereo-capable displays. Anaglyph lets you output a traditional anaglyph red/cyan stereo image for viewing on any display using red/cyan glasses. MV-HEVC lets you output spatial video clips compatible with most VR headsets.

- **Use Constant Bit Rate:** If the Format and Codec you've specified allows you to switch between variable and constant bit rate output, this checkbox lets you force video to render at a constant bit rate.

Optional MP4, H.264, H.265, VP9, or HEVC Controls

If you choose MP4 as the format, or QuickTime with H.264, H.265, or VP9 as the codec, additional options appear, described below. Workstations using NVIDIA GPUs that offer NVENC will present alternative accelerated options, while other workstations offering QuickSync hardware encoding instead will be able to use that option.

- **Use hardware acceleration if available:** DaVinci Resolve supports QuickSync hardware encoding of H.264 and HEVC, if available on your workstation.
- **Quality:** If the currently selected option in the Render to dropdown menu has options for changing the compression quality, this dropdown menu lets you choose the quality you want to use. Otherwise, it's disabled.
- **Restrict to X Kb/s:** You can choose Automatic, or select a maximum data rate with which to export.
- **Encoding Profile:** A dropdown that lets you choose among different encoding profiles, each of which has been optimized for different purposes. The tradeoff is between quality and computational intensity for encoding and playback. The available options are:

Auto: Automatically selects an encoding profile.

Base: For H.264, intended for video conferencing and mobile phone use; highly compressed.

Main: For H.264, intended for SD analog transmission. For H.265, intended for the compression of 4:2:0 video at up to 4K 60fps with a bit depth of 8-bits per channel.

Main10: (H.265 only) Intended for the compression of 4:2:0 video at up to 4K 60fps with a bit depth of 10-bits per channel.

Main 4:2:2 10: (H.265 only) Intended for the compression of 4:2:2 video at up to 4K 60fps with a bit depth of 10-bits per channel.

High: For H.264, intended for Blu-Ray and HD transmission.

- **Entropy Mode:** (called Entropy Coding Mode for compatible Nvidia GPUs) A dropdown that lets you choose which algorithm the encoder should use for compression. The choices are:

CALVC (context-adaptive variable-length encoding): A lower-quality algorithm that's less computationally intensive to process and play.

CABAC (context-based adaptive binary arithmetic coding): A higher-quality algorithm that yields better visual quality at lower bandwidth, at the cost of being more computationally expensive to process and play.

- **Multi-pass encode:** (Available for QuickTime H.264 and H.265) You can choose between Single and Multi-pass encoding. Single pass is faster, but multi-pass yields superior results when quality is important. When you enable Multi-pass, the number of passes performed is automatic.
- **Key Frames:** (Available for QuickTime H.264 and H.265) You can choose Automatic, or select a duration for manual keyframe insertion.
- **Frame Reordering:** (Available for QuickTime H.264 and H.265) On by default, Frame Reordering enables the encoding of B frames to improve the quality of the resulting compressed movie file. Turning off Frame Reordering will speed encoding performance at the expense of visual quality.
- **Rate Control:** (Available for compatible NVIDIA GPUs) Provides six options for controlling Encoding Profile and Entropy Mode.
- **Lookahead:** (Available for compatible NVIDIA GPUs) Lets you specify how many frames for the encoder to examine in advance of compression.

Optional DCP and IMF Controls

If you choose DCP or IMF as the Format, additional options appear, described below.

- **Use interop packaging:** (DCP only, located under Type parameter) Lets you create an Interop DCP package, based on an earlier standard of DCP delivery that is not forward compatible with SMPTE DCP packages.
- **Package Type:** (IMF) Defaults to App2 Extended (App2e), for encoding JPEG 2000 up to 4K.
- **Bit Depth:** (IMF) The bit depth of the encoded IMF video.
- **Encoding Profile:** (IMF) A dropdown that lets you choose among Auto, IMF, and Broadcast.
- **Encoding Level:** (IMF) Provides different choices based on what is selected in Encoding Profile.
- **Maximum bit rate:** (DCP, IMF) Lets you choose how much to compress the result.
- **Lossless Compression:** (IMF) Lets you choose to encode using lossless compression.
- **Slope-Rate Control:** (DCP, IMF) A checkbox lets you specify lossless compression.
- **QStep:** (DCP, IMF) Lets you choose either automatic or manually specified DCP quantization levels at which to compress the video signal when using the Kakadu JPEG 2000 encoder.

Advanced Controls

An advanced settings disclosure button hides the following additional controls, by default.

- **Pixel aspect ratio:** Lets you override the Project Settings and change the PAR of the rendered output to either Square or Cinemascope.
- **Data levels:** Defaults to “Auto,” which simply renders all clips with the data level appropriate to the currently selected codec in the “Render to” dropdown menu, which is usually the preferred behavior. Choosing one of the other options (“Video” or “Full”) outputs all clips using the selected data range. For more information, see *Chapter 9, “Data Levels, Color Management, and ACES.”*
Retain sub-black and super-white data: Turning this checkbox on lets you choose to output media files that preserve overshoots and undershoots, data that’s above the maximum and minimum data levels of the data level you’ve selected, assuming this is supported by the video format and codec you’re exporting to. Otherwise, DaVinci Resolve clips these “out-of-bounds” parts of the signal in an effort to keep your deliverables from violating whatever QC standards you’re adhering to in your grade.
- **Color Space Tag:** A dropdown menu that lets you choose a color space to embed as metadata in the rendered file. This setting defaults to Same as Project if your project’s color science is set to DaVinci YRGB Color Managed, or the ACES Output Device Transform if your color science is set to ACEScct or ACEScct. You can override this option manually from the choices on the dropdown menu.

- **Gamma Tag:** A dropdown menu that lets you choose the gamma to embed as metadata in the rendered file. This setting defaults to Same as Project if your project's color science is set to DaVinci YRGB Color Managed, or you can override this option manually from the choices on the dropdown menu.
- **Data burn-in:** A dropdown menu that defaults to "Same as Project," which leaves the current Data Burn In palette settings enabled while rendering, inserting a window burn into the media being output. Choosing "None" disables window burns while rendering. Note that when rendering as Individual Source Clips, individual clip burn in presets can be assigned if they've been created in the Data Burn In palette.
- **Bypass re-encode when possible:** Turning this checkbox on makes it possible to do a direct copy of the video essence of video items in the Timeline, directly from the source media to the file being output, when the selected Format, Codec, and Type matches the source. This also preserves Alpha channel data for compatible formats.

Bypass re-encode eliminates the need to re-encode video media, preserves quality, and speeds up the output process dramatically, but it only works for clips in the Timeline to which no additional effects have been added. Doing any grading, adding a Resolve FX plugin, adding any overlapping effects or compositing to clips in the Timeline, resizing or stabilizing clips or altering the output sizing of the Timeline, and adding Fusion effects will all necessitate re-encoding the entire clip in order to process these effects. Transitions will require processing but only for the duration of each transition.

There are many situations where this is valuable:

Fast output of simple edits: You've edited a simple cuts-only promo using footage cut from a previously rendered program using QuickTime ProResHQ 422 media, and you're exporting to the exact same format. You can output all of the media very quickly using Bypass re-encode when possible.

Fast output of previously output timelines with small changes: You need to replace a few shots in an effects-intensive program that's already been output. You can import the media file that was output into a new timeline, replace only the required shots with new media. DaVinci Resolve will do a direct copy of all previously rendered media, while re-encoding only the new clips with whatever effects and grading they contain. This lets you quickly re-output a high-quality master file, while preventing you from needing to re-render the entire program.

Fast output of previously output timelines with new audio mixes: You've placed a previously rendered Video+Audio clip onto a timeline and edited a new audio mix clip to replace the old audio mix. In this situation, a new Video+Audio file will be quickly written with the new audio, but the video component of that file won't be re-encoded, again resulting in a fast export at the highest quality.

TIP: For a list of which video formats are compatible with Bypass Re-encode on macOS, Windows, and Linux, as well as which formats are compatible with Alpha channels, see the "Supported Codec List" at the DaVinci Resolve Support page located at: <https://www.blackmagicdesign.com/support/family/davinci-resolve-and-fusion>.

- **Use optimized media:** When this checkbox is turned on, DaVinci Resolve will use optimized media, when available, to do the final render, to save time. If your media has been optimized to the same format as the one you're outputting to (or better), this is convenient. However, if you've optimized to a lower quality format than what you're outputting to, you should turn this checkbox off to force DaVinci Resolve to process all clips using the original media, guaranteeing the best quality available.

- **Use proxy media:** When this checkbox is turned on, DaVinci Resolve will use the generated proxy media, when available, to do the final render, to save time. However, if your proxies are a lower-quality format than what you're outputting to, you should turn this checkbox off to force DaVinci Resolve to process all clips using the original media, guaranteeing the best quality available.
- **Use render cached images:** When this checkbox is turned on, DaVinci Resolve will write media from the cache to the files being output to save time. If you're caching using the same media format you're outputting to (or better), this can be convenient. However, if you're caching in a lower-quality format than the one you're outputting to, you'll want to turn this checkbox off to force DaVinci Resolve to process all media as it's being rendered, writing at the maximum quality you're outputting to.
- **Force sizing to highest quality:** If you've been working with the "When resizing and scaling:" option set to Bilinear to improve performance when working on slower workstations, turning this checkbox on automatically renders all clips using the "Uses Sharper filter" setting of the Image Scaling panel in the Project Settings.

For more information, see *Chapter 4, "System and User Preferences."*

- **Force debayer res to highest quality:** When rendering camera raw media formats that allow variable quality debayering, it's common to lower the debayering quality to improve real time performance while grading. Turning this checkbox on guarantees that media will always be rendered at the highest available quality, saving you from forgetting to manually change the debayer setting back when setting up a render at 3am.
- **Render All Video Tracks:** (When rendering Single Clips) You can un-check Render All Video Tracks and select specific tracks in the Disable dropdown menu to exclude those tracks from the final output on a per render job basis.
- **Enable flat pass:** Three options let you choose whether or not to render each clip with its grade applied.

Off: DaVinci Resolve always applies each clip's grade when rendering.

With clip settings: For each version of a clip, the system will check that version's pass flat flag. If it's turned on, the system disables color correction for that version of the clip. Otherwise, that version will be rendered with its grade intact. Versions can be individually flagged by right-clicking a clip's thumbnail in the Timeline, choosing the submenu of the version you want to flag, and choosing Enable Flat Pass.

Always On: When checked, DaVinci Resolve disables the grade of every clip being rendered.

- **Disable sizing and blanking output:** When turned off, Output Blanking to create letterboxing or pillarboxing is "baked" into the output, as are all sizing adjustments made on the Cut, Edit, and Color pages, including Image stabilization.

When turned on, Output Blanking, Cut and Edit page sizing adjustments, Color page Input and Output Sizing, and Image Stabilization are disabled. Rendered media is rendered either at the source resolution if "Render at source resolution" is enabled in individual clips mode, or to the currently specified resolution of the Timeline or project. If you're outputting via Final Cut Pro or Premiere Pro XML, or Avid AAF, sizing adjustments are output to the XML or AAF files that are created for purposes of round-tripping these adjustments as editable metadata back to an NLE.

Be aware that "Disable sizing and blanking output" does not disable any transform operations that happen within the Fusion page, nor does it disable transforms happening as a result of an OpenFX or ResolveFX plugin applied to one or more clips in the Cut, Edit, or Color pages. All of these effects will continue to be rendered into the final output.

- **Trigger script at:** You now have the option of triggering a script to execute before or after rendering a timeline, by checking the “Trigger script at” box.

Start: Executes the script before the render job.

End: Executes the script after the render job.

Script: Chooses the specific script to run. You can select the specific script to execute using the corresponding dropdown menu. Scripts must be written for the Resolve scripting framework in either Python or Lua, and placed in the following directory:

MacOS: /Library/Application Support/Blackmagic Design/DaVinci Resolve/Fusion/Scripts/Deliver/

Windows: C:\ProgramData\Blackmagic Design\DaVinci Resolve\Fusion\Scripts\Deliver

Linux: /opt/resolve/Fusion/Scripts/Deliver

- **Render full extents:** (When rendering Individual Clips) When this box is checked, the entire original media clip is rendered out completely, instead of just the portion delineated by the In and Out points of the timeline clip.
- **Add X frame handles:** (When rendering Individual Clips) Lets you specify front and rear handles to be output in frames. This is particularly useful in round trips, when the finishing editor might want additional handles with which to roll edit points or add transitions while fine-tuning the graded edit.
- **Tone Mapping:** (Available in Single clip mode if Dolby Vision or HDR10+ is enabled in Project Settings) When set to None, the timeline is output using the current color management settings. When set to either Dolby Vision or HDR10+, you can choose to output the timeline at a specific peak nit level, color space, gamma, and Data Level using either the Dolby Vision or HDR10+ metadata available to guide the tone mapping operation you’ve selected. This makes it easy to set up multiple jobs to output HDR outputs at varying levels, as necessary.

DCP and IMF Composition Settings

If you’ve selected either DCP or IMF from the Format, a Composition Settings group appears with the following parameters when you click the disclosure control, which let you populate generic DCP and IMF composition metadata, as well as various studio specific settings:

- **Composition name:** The name of the exported composition.
- **Issuer:** The organization providing the composition.
- **Use current date:** A checkbox that lets the current date be used as the Issue date automatically.
- **Issue date:** The date the composition is issued.
- **Content kind:** A dropdown provides a list of acceptable choices for defining the content.
- **Content originator:** A text field to add who is responsible for this content.
- **Include volume index and output profile list:** Check this box to include these xml files in the package.
- **Content version label:** Meant to identify the version of the content being provided.
- **Annotate xml using composition name:** Auto-populates Asset Map, Composition Playlist, and Packing List with data from the project. Otherwise these three fields are manually editable.
- **Annotate reel index as suffix (DCP only):** Auto-populates Reel Annotation with data from the project. Otherwise this is manually editable.
- **Annotate media using filename:** Auto-populates Main Video Track and Audio Track 1 with data from the project. Otherwise these three fields are manually editable.

Subtitle Controls

The Subtitle Settings group exposes controls governing how to export subtitles in your program:

- **Export Subtitle checkbox:** Lets you enable or disable subtitle/closed caption output.
- **Format pop-up:** Provides four options for outputting subtitles/closed captions.
 - As a separate file:** Outputs each subtitle track you select as a separate file using the format specified by the Export As pop-up. A set of checkboxes lets you choose which subtitle tracks you want to output.
 - Burn into video:** Renders all video with the currently selected subtitle track burned into the video.
 - As embedded captions:** Outputs the currently selected subtitle track as an embedded metadata layer within supported media formats. There is currently support for CEA-608 closed captions within MXF OP1A and QuickTime files. You can choose the subtitle format from the Codec pop-up that appears.
- **Export As:** (Only available when Format is set to “As a separate file.”) Lets you choose the subtitle/closed captioning format to output to. Options include: IMSC1, DFXP, SRT, and WebVTT.
- **Include the following subtitle tracks in the export:** (Only available when Format is set to “As a separate file.”) A series of checkboxes lets you turn on which subtitle tracks to output.
- **Codec:** (Only available when Format is set to “As embedded captions.”) Lets you choose how to format embedded closed captions; choices include: Text and CEA-608.

NOTE: Neither analog (Line 21) nor digital (CEA-708) closed caption output via Decklink or UltraStudio is supported at this time.

Audio Panel

This panel contains all audio-oriented parameters.

- **Export audio:** Turn this checkbox on to render the source audio, or audio that you’ve synchronized in DaVinci Resolve, along with the media being output by DaVinci Resolve.
- **Format:** You can choose which format of audio you want to render. Depending on which format you choose, different audio codec options will appear below.
 - MXF OP1A(IMF):** Generates media files that conform to the OP1a (Operational Pattern 1a) SMPTE 378M format of MXF media for file exchange.
 - MXF OP-Atom:** Generates media files that conform to the SMPTE 390M format of MXF media for file exchange.
 - QuickTime:** Exposes all available formats of audio compatible with QuickTime.
 - MP4:** Generates media in the MPEG-4 file format.
 - WAVE:** Generates media in the WAVE file format.
 - MP3:** Generates media in the MPEG-3 file format.
 - FLAC:** Generates media in the FLAC file format.
- **Codec:** Lets you choose between Linear PCM (the default), AAC audio, IEEE Float, or MP3. AAC audio encoding is only available on macOS.
- **Sample Rate:** Lets you set the sample rate, in Hz, of the output of the audio.
- **Bit Rate Strategy:** (Available for AAC encoding) You can choose among Constant Bit Rate, Average Bit Rate, Variable Bit Rate Constrained, and Variable Bit Rate.

- **Quality:** (Available for AAC encoding when Bit Rate Strategy is set to Variable Bit Rate) Five settings you trade between speed and quality when encoding AAC audio.
- **Data Rate:** (Available for AAC encoding when Bit Rate Strategy is set to Constant, Variable, or Variable Constrained) Lets you choose the maximum data rate for AAC encoding.
- **Bit Depth:** Lets you specify the bit depth at which to output the source audio.
- **Render one track per channel checkbox:** This checkbox lets you specify whether you want to export each channel as an individual track in the structure of the exported file. Multi-channel formats can be specified (2.0, 5.1, etc.), or can be output as flat multi-mono files.
- **Render as discrete audio tracks checkbox:** This checkbox is similar to the above, in that it still outputs one channel per track. However, discrete tracks have no positioning info, nor do they group as multi-channel clips (2.0, 5.1, etc.). For example, a mono file exported with Render one track per channel, comes in front and centered. The same file exported as a discrete audio track has no built in position information.
- **Output Track #:** This dropdown menu lets you choose which Main or Submix bus you want to output. A Plus button to the right lets you add additional tracks you want to export in this job, so if you want to export multiple Mains or multiple Subs, you can add more Track pop-ups.

When you choose a track with multiple channels of audio, a field appears showing how many channels will be output; you have the option of using a virtual slider to change how many channels are output.

In Single Clip Mode, when you choose “All Timeline Tracks, each audio track in the current timeline will be rendered as an individual track in the rendered media file you’re outputting. Multi-channel tracks containing multi-channel clips will output audio tracks containing the same number of channels in the output media, which means you can output mixed combinations of mono, stereo, 5.1 or 7.1, and adaptive audio tracks, and each separately rendered audio track will reflect identical channel mappings.

- **Language:** (Only available when outputting to the IMF format) This dropdown menu lets you choose the language of a particular output track when you’re outputting to IMF. Since IMF files can accommodate multiple audio tracks containing different mixes for different regions, it’s important to identify each output track you’re outputting by language.
- **Content:** (Only available when outputting to the IMF format) This dropdown menu lets you choose the content of a particular output track when you’re outputting to IMF. Since IMF files can accommodate multiple audio tracks containing different mixes and content, it’s important to identify each output track you’re outputting by content.
- **Audio Normalization:** You can select loudness standard presets from the Audio tab in the Render settings on the Delivery page to automatically have the deliverable normalized accordingly. These controls let you control the audio normalization on a per output basis. Unchecking this box bypasses audio normalization and optimization.

Normalize to Standard: Check this box to normalize loudness or peak (scaling whichever first exceeds thresholds) to a standard.

Optimize to Standard: Check this box to optimize loudness and peak (ensuring that loudness meets the desired range and peaks remain lower than threshold) to a standard.

Standard: Choose the normalization standard you want to apply.

Target Level: Set the normalization level in dBFS.

Output Track #: Check the box to apply the normalization to these outputs.

File

This panel contains all other parameters.

- **Filename Uses:** Three options let you automatically name the media file(s) that are output automatically.

Use Custom Filename: Lets you enter your own name in the Custom name/File prefix field.

Use Timeline Name: (When rendering a Single Clip) When this option is selected, the name of the Timeline is used.

Use Source Filename: (When rendering Individual Clips) When this option is selected, the filename of each clip's corresponding source media file is cloned, and used as the filename of media being output by DaVinci Resolve. This is preferred when you're generating offline media for use by an editor that you later want to reconfirm to the originating DaVinci Resolve project. When this checkbox is turned off, you can customize filenames using the other options in this section of settings.

- **Custom name:** Lets you use custom text to name all rendered files. If you're not using the source filename, and not rendering to a file format that uses timecode, you can enter a filename here. When editing the Custom Name or File Prefix (or File Suffix), you can use "metadata variables" that you can add as graphical tags that let you display clip metadata. This is especially useful when rendering Individual Source Clips.

For example, you could add the corresponding metadata variable tags %scene_%shot_%take and the File Prefix would be written as "12_A_3" if "scene 12," "shot A," "take 3" were in the source clip's metadata.

For more information on the use of variables, as well as a list of all variables that are available in DaVinci Resolve, see *Chapter 16, "Using Variables and Keywords."*

- **File suffix:** Lets you add custom text and/or metadata variables (described previously) to the end of all rendered files.
- **Use unique filenames:** (When rendering Individual Clips, and only when Filename uses is set to Custom name) When enabled, additional characters are added to every rendered media file to guarantee that each rendered media file has a completely independent name. This prevents multiple rendered clips from the same source media file from overwriting one another when saved to the same directory. "Uniquely" named clips append the clip name with the track and clip number identifying a clip's position in the currently selected session. For example, a clip that's linked to a media file named "DropThatThingCU.mov," and edited as the twenty-fifth clip on track V2, will be named "DropThatThingCU_V2-0025.mov" when rendered. When enabled, two other options are revealed.

Use unique filename prefix/Use unique filename suffix: (When Use Unique Filenames is on) Radio buttons let you choose whether to add the unique identifier at the beginning or end of a clip. Choosing Prefix would result in "V2-0025_DropThatThingCU.mov," whereas choosing Suffix would result in "DropThatThingCU_V2-0025.mov" when rendered.

- **Add source frame count to filename:** (When rendering Individual Clips, and only when Filename uses is set to Custom name) When enabled, the source frame number of each clip is appended to the end of the rendered file name. This is another way to make sure that multiple rendered clips with custom names don't overwrite one another.
- **Use filename digits:** Lets you specify how many digits to use when rendering an image sequence, although the specified digits will also be used for any media format. This is particularly useful if you're outputting media to be used by an application that has strict requirements for image sequence numbering. Defaults to eight digits.

- **Each clips starts at frame:** (When rendering Individual Clips) This permits timecode to be written to the header, and frame count to be written to the filename of the image sequences, which is ideal for VFX workflows.
- **Start timeline timecode at:** (When rendering Single Clip) This option is only available when rendering clips in Single clip order. Specifies the timecode that will be written to the media being output by DaVinci Resolve. For DPX files, timecode is written into the header data, and is simultaneously converted to a frame count that's inserted into the filename of each frame file, which provides a logical count of the frame numbers. For other media formats, timecode is written to the appropriate metadata container. You may find it useful to use custom start times, for example starting each reel of a project at a particular value, depending on the standards employed at your shop.
- **Place clips in separate folders:** (When rendering Individual Clips) Useful if you need to preserve the filenames of files you're outputting when the filenames of clips coming from the same source media file may cause them to overwrite one another. This option is also commonly used when rendering VFX shots for additional post-production work, allowing the VFX department to identify clips quickly and distribute the work accordingly.
- **Preserve Source directory levels:** (When rendering Individual Clips) Retains a user-specified depth of the original directory structure used by a clip's corresponding source media file, recreating it when rendering new files for output. The number you select determines how many levels of subdirectories DaVinci Resolve will automatically create within the currently specified "Render job to" directory to match the path used by the source files. Defaults to 0, which creates no matching subdirectories.

After Head/From Tail: When setting how many directory levels of each clip's file path to preserve (using the "Preserve x levels" parameter), click one of these buttons to specify whether that number of path levels is defined relative to the head or the tail of each media file path.

Preserved Path: Shows you a preview of the preserved path you've set up so you know you've gotten it right.

- **File Subfolder:** (Only appears in Additional Output panels) Lets you specify a subdirectory into which to render the media files being output. If the specified subdirectory doesn't exist, a new one with that name will be created within the currently specified "Render job to" directory.
- **Use commercial workflow:** (When rendering Individual Clips) Automatically renders every version that's applied to each clip in the session, except for versions that have been flagged using the "Render Disabled" flag, found in the Version submenu for each clip in the Timeline. This option is typically used when you've graded multiple versions of a clip to be used for VFX work, and you want to deliver each grade as a separate media file. This is also used when rendering programs for commercial broadcast where you have two or more versions of a grade for each scene. When using this option, alternate methods of outputting each rendered media file are used, and four additional settings are revealed.

Alternative pass offset: Lets you separate the timecode values written into each version of a clip with an offset. For example, if the default version timecode is 01:00:20:00, and you select a 10 minute offset in the Alternative Pass Offset timecode entry, then the second graded version of that clip will start at 01:10:20:00, the third version will start at 01:20:20:00, and so on until every version is rendered. You can offset the clips by whatever value you like, but the idea is to make it easy for editors and VFX artists to find the versions of each grade. If the clips are shared with a finishing artist, and they know that each alternate pass is 10 minutes apart, then it's easy for the finisher to change the clip version just by adding 10 minutes to the referenced timecode. To simplify the workflow further, you can put separate source reels in separate folders using the next three options.

Place reels in separate folders: Automatically places all media that's output using a particular reel name into corresponding folders.

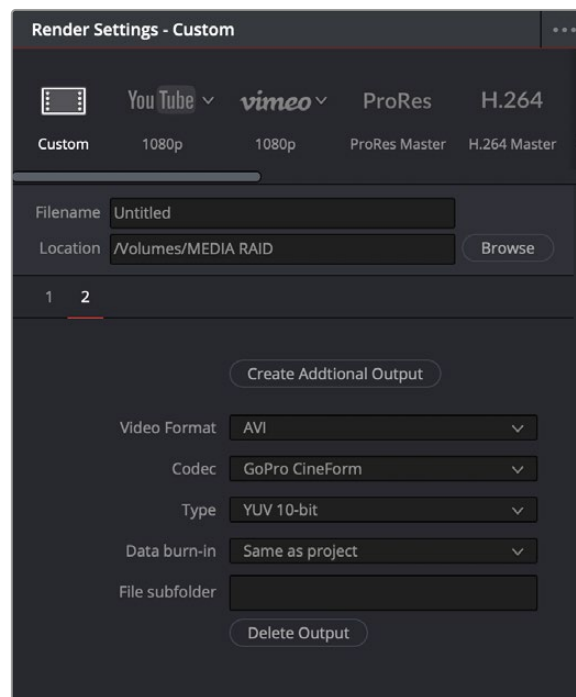
Place clips in separate folders: Automatically places alternate grades of clips into separate folders.

Use version name for folders: Labels each folder with the name of the version when using the Commercial Workflow option.

- **Render speed:** A dropdown menu lets you throttle the speed at which media is rendered. Ordinarily you'll leave this set to the default of Maximum. However, some storage systems that are shared by multiple rooms in a facility use storage area networks (SANs) with insufficient bandwidth for multiple real-time image streams. DaVinci Resolve's incredibly fast rendering speeds can cause playback problems with other users accessing the SAN if available bandwidth is insufficient. In this case, you can throttle the render speed to limit SAN bandwidth usage to between 1 to 50 percent of full rendering speed.
- **Disk space currently used:** Shows the amount of disk space available on the target volume.
- **Disk space used after render:** Shows the new disk usage based on the specified range of the current session that you're rendering.

Additional Outputs

Each job you create in the Render Settings defaults to a single output. However, you can create multiple outputs when you need to deliver multiple versions of media, with individual video formats and codecs and different data burn-in settings, to be rendered into individually named subfolders (optional).



The menu command for creating an additional output, shown next to an existing additional output in panel 2

This can be useful for setting up multiple rendered passes when your client requires two sets of media, for example QuickTime ProRes 422 (HQ) media along with MXF DNxHD media. This is also useful when you need to output two sets of media where one has window burns, and the other is clean.

To add additional outputs in the Render Settings:

- Choose Create Additional Output from the Render Settings Option menu. A row of numbers below the Filename and Location controls let you open each output you create and adjust its settings. You can have as many outputs as you require.

To remove an additional output:

- Open the additional output panel you want to remove, and click the Delete button at the bottom.

How to Avoid Overwriting Clips When Rendering Output Media

Three of the options described previously, “Use unique filenames,” “Place clips in separate folders,” and “Use commercial workflow,” are all ways of organizing your rendered media to avoid overwriting rendered clips that happen to share the same file name. These options are necessary because each clip has one logical range of timecode, and because multiple clips often refer to a single source media file with one name.

When rendering a clip, DaVinci Resolve automatically overwrites any other media files that have the same name. So, in instances where you’re trying to preserve the previous filename of the source media file, or where you’re rendering out multiple versions of the same clip, it’s entirely likely that the clips you’re trying to output will overwrite one another, leaving you with the last clip you rendered. The three options mentioned previously prevent this in different ways.

Defining a Range of Clips and Versions to Render

Once you’ve defined the Render Settings, now you need to decide how much of the Timeline you need to render. A Mini-Timeline and Thumbnail timeline are available to help you navigate your project’s clips in order to choose which ones you want to render. Track controls let you enable and disable whole tracks from being output; for example if you need to render a textless version of a timeline in which all the title clips are on track V4, you can disable track V4. Furthermore, you can also use these controls to choose which clip versions you want to render.

To render the entire Timeline:

- Choose Entire Timeline from the Render dropdown in the Deliver page timeline. This option only appears if clips are not filtered.

To disable a video or audio track to exclude those clips from being rendered:

- Click the Video or Audio Disable Track button for the tracks you want to exclude.

To render a filtered subset of clips in the Timeline:

- 1 Open the Color timeline, if it’s not already shown, and choose an option from the Timeline Filter dropdown to the right of the Clips button in the Interface toolbar.

The contents of the Thumbnail timeline are restricted to show only the clips matching the selected criteria. For example, if you’ve already rendered a session, but you’ve since made some changes, you can use one of the “Show Modified Clips” options to display only the clips that have changed within a particular timeframe. Another possibility is to choose the “Show Unrendered Clips” option to show all clips that have not yet been rendered.

- 2 Choose “All Filtered Clips” from the Render dropdown in the Timeline toolbar.

To clear clip filtering:

- Choose All Clips from the Timeline Filter dropdown to the right of the Clips button in the Interface toolbar.

To define a continuous range of clips to render:

- 1 To define the first clip in the range you want to render, do one of the following:
 - Right-click a clip thumbnail in the Thumbnail timeline and choose Mark In.
 - Position the playhead in the Timeline or the Viewer, and press the I key, or right-click the Timeline ruler and choose Mark In.
- 2 To define the last clip in the range you want to render, do one of the following:
 - Right-click a clip thumbnail (in the Color timeline) or a clip (in the Edit timeline) in the Thumbnail timeline and choose Mark Out.
 - Position the playhead in the Timeline or the Viewer, and press the O key, or right-click the Timeline ruler and choose Mark Out.

In and Out points appear within the Timeline ruler, and an orange bar shows the range you've selected to render. The In and Out fields update with the first and last frame numbers, in timecode and frame count, and the Duration field updates with the total number of frames you'll be rendering.

IMPORTANT: If you're in Individual Clips mode, In and Out points automatically snap to the nearest clip In or Out point in the Timeline. You cannot render partial clips in Individual Clips mode, but you may do so in Single Clip mode.

To render a single clip:

- Open the Thumbnail timeline if it's not open already, Right-click any clip thumbnail, and choose Render This Clip.
- An orange bar in the Timeline ruler shows that clip is selected for rendering. If you need to render several clips individually, you can select each clip one at a time to add as individual jobs to the Render Queue.

Choosing Which Versions to Render for Each Clip

By default, the currently selected version that was set in the Color page is rendered for each clip. If you want to render a different version, the easiest thing to do is to make sure they're selected on the Color page Timeline before you open the Deliver page.

However, a Versions submenu, within the Thumbnail timeline's contextual menu for each thumbnail, also provides options to manage grade versions. These commands are duplicates of options that are available from the Thumbnail timeline of the Color page.

To choose which version to render:

- Right-click any clip thumbnail in the Thumbnail timeline, and choose a version from the Versions submenu.

TIP: You can right-click a clip in the Thumbnail timeline of the Color or Render screen and rename any version of a grade. This can assist a facility's workflow when sharing material between suites and applications.

Using the Render Queue

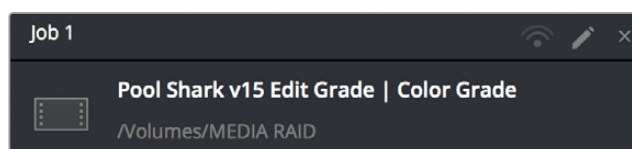
Once you've defined the settings necessary to render the type of media you require, and the range of the current session you want to render, you need to add all that information as a job to the Render Queue. You can add as many jobs to the Render Queue as you need, depending on what files you need to output.

Each job can have individually specified ranges of clips and individual clip settings, which can include different render directories, different formats, resolutions, data levels, burn-in settings, and so on. As a result, you can use the Render Queue to queue up the render of multiple sections of the current session, or multiple versions of the same media. Furthermore, you can queue up multiple sessions, if you have several differently graded sessions.

To add a job to the Render Queue:

- 1 Select a timeline.
- 2 Choose the settings you require in the Render Settings, using one of the Presets, or by choosing your own custom settings.
- 3 Choose a range of clips to render using the Deliver page Timeline using the procedures described in the previous section.
- 4 Click the Add to Render Queue button at the bottom of the Render Settings.
- 5 If you haven't chosen a location for the render yet, you'll be prompted to do so now via a File Destination dialog, so choose a location and click Ok. If there's already media in the render location you've specified, you may also see a dialog telling you "This render may overwrite existing clips in this folder." If you want to continue, click Yes, otherwise click No.

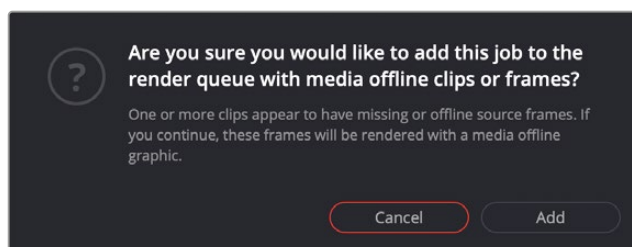
That render setup is now added as a job to the Render Queue, showing the project and timeline name, and location path where the render will be written to.



A selected job in the Render Queue

Media Offline Warning in Render Queue

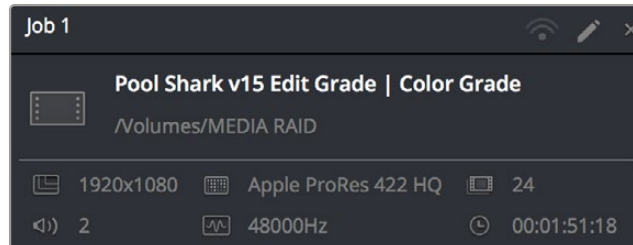
When you attempt to add a job to the Render Queue with a timeline that contains any offline material, DaVinci Resolve automatically gives you a warning. You may choose to either cancel adding the job, or to add it anyway, knowing you're about to render one or more offline clips.



Media Offline warning box that appears if your timeline contains offline clips or frames and is added to the Render Queue

To show more details about jobs in the Render Queue:

Click the Render Queue Option menu (at the upper right-hand corner) and choose Show Job Details. Each job now lists the frame size, format, frame rate, audio channels and sample rate, and duration below the name and location path.



A selected job in the Render Queue with job details shown

To rename a job:

Jobs can be given custom names simply by clicking on the default job name (Job 1, Job 2, and so on) and typing a new name of your own. This can be useful for setting up jobs that you may be re-rendering over and over as you work on a project.

To start rendering:

- 1 If you want to restrict rendering to only selected jobs in the Render Queue, then select one or more jobs by clicking on one, and then Command-clicking on others to choose discontinuous jobs, or Shift-clicking on another to select an entire range of jobs. When you select one or more jobs, only the selected jobs will be rendered. If no jobs are selected, then all jobs in the queue will be rendered.
- 2 Click the Start Render button, underneath the Viewer to the right of the interface.
- 3 If there are jobs in the Render Queue that have already been rendered, a dialog will appear asking “Selected items contain already rendered items. Do you want to re-render them?” Clicking Yes will re-render all jobs in the Render Queue. Clicking No only renders the jobs that have not yet been rendered. Clicking Cancel cancels the entire rendering operation.

Rendering begins, starting with the highest job in the list. The Overall Progress bar starts to fill up, from right to left, indicating how much of what’s been queued up has been rendered so far. While rendering is in progress, the Start Render button changes to the Stop Render button, which can be clicked at any time to halt rendering.

TIP: While rendering is in progress, a small progress bar will appear on the DaVinci Resolve icon in the dock of Mac OS X, or on the taskbar of Windows.

To remove jobs from the Render Queue, do one of the following:

- **To clear a specific job:** Click the X at the upper right-hand corner of a job’s entry in the Render Queue.
- **To clear all previously rendered jobs:** Click the Render Queue Option menu (at the upper right-hand corner) and choose Clear Rendered.
- **To clear all jobs:** Click the Render Queue Option menu (at the upper right-hand corner) and choose Clear All.

To change a job that has been rendered to appear unrendered again:

Right-click any rendered job, and choose Clear Render Status. You can also select multiple jobs to change their rendered status all at once. This makes it easy to re-render the exact same job.

To edit a job that has or has not been rendered:

- 1 Click the Pencil button in the Render Queue to select it.



Clicking the pencil icon to edit a job in the Render Queue

The selected Render Queue's settings repopulate the Render Settings list, and resets the selected range of the Timeline corresponding to that job.

- 2 Change whichever settings you need to.
- 3 When you're finished editing the job, click the Update Job button that appears at the bottom of the Render Settings, or you can click Add New Job to create a new job with the changes you've made, leaving the previous job untouched.

NOTE: If you click the Pencil button again without clicking Update Job, you'll be prompted to Save, Cancel, or Don't Save.

To review clips that correspond to rendered jobs:

- **To show a rendered clip in the Media Storage browser:** Right-click any rendered job, and choose Reveal in Media Storage.
- **To show a rendered clip in your computer's file system (Mac OS only):** Right-click any rendered job, and choose Reveal in Finder.

Rendering Jobs from Multiple Projects at Once

You can also add multiple projects from the currently open PostgreSQL or local project library to the render queue all at once. This can be exceptionally useful in situations where you've broken a program into multiple reels, with each reel being a different project.

To render output from multiple projects at once:

- 1 Open each project, set up whatever jobs you want to render in the Render Queue, and save that project without rendering.
- 2 When you've set up the last project, click the Render Queue Option menu (at the upper right-hand corner) and choose Show All Projects.

All queued items in projects belonging to the currently selected user (if using a network project library) or in the currently specified disk location (if using a local project library) now appear in the Render Queue.

- 3 Click Start Render to begin rendering every job from every project in the queue.
- 4 When you're finished, turn Show All Projects off to go back to displaying only the current project's render queue items.

Remote Rendering

If you have multiple DaVinci Resolve workstations on the same network, you can send a job in the Render Queue from the workstation you're using (referred to as the "artist workstation") to one of the "remote workstations" on the network using remote rendering. This lets you use any one of your currently unused secondary workstations to render your jobs, while you continue working on your main workstation.

In order to use remote rendering, you must adhere to the following three criteria:

- Both the artist workstation and the remote workstation must have DaVinci Resolve Studio installed. Remote rendering does not work with the free version of DaVinci Resolve.
- Both the artist workstation and the remote workstation can use the same shared network project library, or any other Postgres project library that is connected to either one of the machines, or to a dedicated Remote Project Library Server.

For more information on setting up and using shared project libraries, see *Chapter 196, "Managing Project Libraries and Project Servers."*

- Both the artist workstation and the remote workstation must have access to the same media files on either the same storage volumes, or identically named storage volumes.

Using Multiple Project Libraries in Remote Rendering

You can set up remote renders for projects in all connected DaVinci Resolve project libraries, not just the currently active one. To activate this feature, check the "Automatically scan other project libraries for Remote Rendering jobs" box in the General settings of the System tab, of the DaVinci Resolve Preferences.

Sharing Storage

It's important that both the artist and remote workstations have access to the same media on the same named storage volume for remote rendering to work properly. This can be done via some manner of shared storage, such as a SAN. However, it can also be done by mounting the same volume over your network. This will be slower, but it will work.

If you're mixing Mac OS X, Windows, and Linux workstations for remote rendering, you'll need to use the Mapped Mount column of the Media Storage Locations list in the Media Storage panel of the System Preferences to add each volume's path as it's understood on the workstation it's attached to. For example, on a Windows workstation that's accessing volumes from a Linux workstation, type in the Linux-style file paths in the Mapped Mount column for each scratch disk that's listed.

Setting Up and Using Remote Rendering

Using remote rendering is easy, but it does require a bit of preparation.

- 1 Make sure the storage volume containing the media being referenced by the project you want to render is mounted on both the artist and remote workstations.
- 2 Open DaVinci Resolve on the remote workstation, and do one of the following:
 - When the Project Browser opens, right-click anywhere and choose Remote Rendering.
 - If you've already opened a project in DaVinci Resolve, you can also choose Workspace > Remote Rendering.

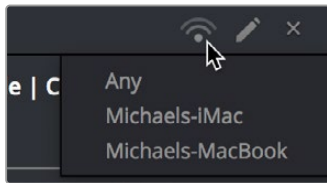
DaVinci Resolve will automatically open to the Deliver page, awaiting jobs to be assigned for automatic rendering.

- 3 On the artist workstation, add a job to the render queue as you normally would.
- 4 Click the Remote Rendering button for that job in the Render Queue and one of the options from the list that appears:

Any: Automatically assigns that job to the next workstation that isn't currently rendering anything. If all remote rendering workstations are rendering, assigns it as the next job in line.

YourComputer.local: The artist workstation with the name "YourComputer." Choose this if you want to render the job locally, and not remotely.

Other Workstations on Network: All other remote rendering workstations are listed below, so you can choose which specific workstation you want to assign a job to.



Clicking the Remote Render button to remotely render a job

- 5 Click Start Render. The job is sent to the remote workstation you selected and is rendered, while you're free to continue working on your artist workstation.

When You're Finished Remote Rendering

Once you're done using a particular DaVinci Resolve workstation in Remote Rendering mode and you want to go back to using it as an artist workstation, choose Workspace > Remote Rendering to exit remote rendering and return to the Project Manager.

Setting Up a "Headless" Remote Rendering Workstation

DaVinci Resolve allows remote rendering clients to operate in a so-called "headless" mode, with no GUI. This can be accomplished from the command line, by opening the directory where the app is located and then running DaVinci Resolve in Remote Rendering (-rr) mode using the correct command line syntax for your operating system. Once run in this way, DaVinci Resolve silently and invisibly waits on that system for remote rendering jobs to be sent to that workstation.

On macOS

Open Terminal.

Change the directory to:

```
cd /Applications//DaVinci\ Resolve/DaVinci\ Resolve.app/  
Contents/MacOS/
```

Run the following command:

```
./Resolve -rr
```

On Windows

Open the Command Prompt.

Change the directory to:

```
C:\Program Files\Blackmagic Design\DaVinci Resolve\
```

Run the following command:

```
Resolve.exe -rr
```

On a Linux CentOS 6.8 system

Open Terminal.

Change the directory to:

```
cd /home/resolve/Cyclone/
```

Run the following command:

```
./script.start -rr
```

On a Linux CentOS 7.x system

Open Terminal.

Change the directory to:

```
cd /opt/resolve/bin
```

Run the following command:

```
./resolve -rr
```